

Markscheme

Mathematics: analysis and approaches
Practice question bank

79. (a)

The height of a plant (in cm) after n weeks follows an arithmetic sequence for the first 10 weeks, with $u_1 = 2$ and common difference 1.5. After week 10, the weekly growth of the plant halves each week (rather than staying constant).

Find the height of the plant at the end of week 13.

height at week 10: $u_{10} = 2 + 9(1.5) = 15.5$ *(M1)* **A1**

weekly growth after week 10: 1.5, 0.75, 0.375, 0.1875, ... (halving each week) *(M1)*

growth added over weeks 11-13: $0.75 + 0.375 + 0.1875$ *(M1)*

height = $15.5 + 1.3125 = 16.8125$ **A1**

[6 marks]